

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE CODE: CSA1710

COURSE NAME: Artificial Intelligence

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

**Time Duration: 1 Hour
Total Marks:30**

QUESTIONS: 5

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I Ravi Teja .V.V (19056503) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



19+19+28+17+12
(90)
FF



Q1 Sports Achievement Award:

Given:

if a player scores above 90%, then the Player Wins a gold medal.

Divya Scored above 90%.

1. Propositional logic

P = Divya scores above 90%.

G = Divya Wins a gold medal

$$P \rightarrow G$$

Goal: G

2. Convert into CNF

The implication is:

$$P \rightarrow G$$

Using:

$$A \rightarrow B \equiv \neg A \vee B$$

$$\therefore, P \rightarrow G \equiv \neg P \vee G$$

$$C1: \neg P \vee G$$

$$C2: P$$

Negate the goal:

Goal

G

Negated goal.

Let P

$\neg G$

C3: $\neg G$

4. Resolution

Resolve C1 & C2

$G \vee G, P$

Since P is $\neg P$ cancel:

G :

New clause:

C4: G

$G, \neg G$

$\therefore$, Empty clause ($\square$) is obtained, the assumption that Divya doesn't win a gold medal leads to a contradiction Divya wins a gold medal. Hence the goal is proved using resolution Algorithm.

Q2: Job Interview Outcome

1. if an applicant passes the interview, then applicant is hired.

2. Teja passed the interview.

Goal: Prove that Teja is hired.

Propositional logic.

Let $P = \text{Teja passes the interview}$

$H = \text{Teja is hired.}$

Knowledge Base:

$P \rightarrow H$

P

Goal:

H

2. CNF Conversion

$P \rightarrow H$

using

$A \rightarrow B \equiv \neg A \vee B$

$\therefore$,

$\neg P \vee H$

clauses

$C1: \neg P \vee H$

$C2: P$

3: Negate the Goal

H

Negated Goal

$\neg H$

$C3: \neg H$

4: Resolution Steps

Resolve $C1$ & $C2$

$\therefore$, H

$C4: H$

Now resolve $C4$ & $C3$

$H, \neg H$

∴, The empty clause is obtained.

Teja is hired.

Hence, the goal is proved using resolution Algorithm.

Q3 - Attendance Credit:

Given:

1. if a student attend all classes, then the student get attendance Credit.

2. meena attended all class.

Goal: Prove that meena gets attendance Credit.

Knowledge Base:

$$A \rightarrow C$$

A

Goal:

C

CNF Conversion:

$$A \rightarrow C \equiv \neg A \vee C$$

∴

$$C_1 = \neg A \vee C$$

$$C_2 = A$$

Negated goal:

$$C_3 = \neg C$$

Resolution

resolve C_1 and C_2

$$(\neg A \vee C), A$$

Now resolve :

$C_1, \neg L$

finally $\square$

The empty clause is derived means getting attendance credit.

Q4: Car rental eligibility.

Given:

L = Swresh has a valid license

R = Swresh can rent a car.

$L \rightarrow R$

$\neg$
Goal

R

CNF Conversion

$L \rightarrow R \equiv \neg L \vee R$

$C_1 = \neg L \vee R$

$C_2 = L$

Negated goal:

$C_3 = \neg R$

Resolution

$(\neg L \vee R), L$

$\therefore, R$ Now, $R, \neg R \therefore$

$\therefore$ $\square$ empty clause is obtained.

Suresh can rent a car. is logically valid using inference and resolution.

Q5: Vehicle registration process:

Knowledge representation: E = car passes emission test
 W = car is road worthy, R = car can be registered.
 $E \rightarrow W$, $W \rightarrow R$, E Goal: R

CNF Conversion:

$$E \rightarrow W, C_1 = \neg E \vee W$$

$$\text{Second: } W \rightarrow R \Rightarrow C_2 = \neg W \vee R$$

fact: $C_3 = E$ Negate goal $C_4 = \neg R$ resolution step 1

resolve C_1 & C_3 : $(\neg E \vee W), E \therefore C_5 = W$

Resolve C_2 & C_5 : $(\neg W \vee R), W \therefore C_6 = R$ This means the car can be registered.

resolution step 3:

$$R, \neg R$$

$\therefore \square$

The empty clause is obtained, proving that the negation of the goal is impossible.